STANDATA variance 23-BCV-003

Building

Insulation below a heated floor

Date Issued: May 2024

Purpose

To provide an alternative solution for providing adequate insulation value below a heated floor in contact with the ground.

Discussion

The objective of insulation below a heated floor in contact with the ground is to reduce the environmental impact due to heat transmission to the ground from the heated floor. Section 9.36. of the National Building Code – 2023 Alberta Edition (NBC(AE)) ensures this objective is achieved. However, this code provision impacts the viability of hydronic heating systems in floors in contact with the ground. Hydronic heat is an energy efficient method for heating a home that uses tubing to run a hot liquid in the floor, along base board heaters or through radiators to heat your home. Also referred to as radiant heating, this type of system has become increasingly popular in Alberta for homeowners who want added comfort and control in their heating zones, savings through lower heating bills, and a decrease in their environmental impact by making smart green building choices.

Current under slab-insulation code provisions do not take into account hydronic heating systems as an appropriate method of adequate insulation value below a heated floor in contact with the ground. Municipal Affairs identified that the provisions for current slab-insulation are more than what is required to meet energy efficiency objectives and is consulting with the National Research Council (NRC) to recommend an appropriate insulation rating that will allow for hydronic heating installations while also supporting Alberta's energy efficiency goals. This province-wide variance will allow an alternative solution to the provisions under Section 9.36. for use in installations below heated floors in contact with the ground. It is important to note, this variance will not remove the insulation requirement from Section 9.36. of NBC(AE), but the variance will allow, as an alternative, less than the RSI 2.84 (R-16) of under-slab insulation where hydronic or radiant floor heating is installed in the province.

Background information

- An energy performance analysis conducted on a "typical" constructed home in Alberta demonstrated a reduction in energy consumption when an RSI 0.88 (R 5) insulation below a heated floor is increased to an RSI 1.32 (R 7.5) insulation, providing a 30% increase in thermal performance. In the event a project is aiming for a high-performance energy target, the energy performance analysis demonstrated that increasing the insulation below a heated floor above RSI 1.32 (R 7.5) has marginal impacts on comfort and total annual energy consumption.

Code References

Sentence 9.36.2.8.(1)

9.36.2.8. Thermal Characteristics of Building Assemblies Below-Grade or in Contact with the Ground

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1) Except as provided in Sentence (2) and Article 9.36.2.5., the effective thermal resistance of *building* assemblies that are below-*grade* or in contact with the ground shall be not less than that shown for the applicable heating-degree day category in
a) Table 9.36.2.8.-A, where the ventilation system does not include heat-recovery equipment, or
b) Table 9.36.2.8.-B, where the ventilation system includes heat-recovery equipment conforming to Article 9.36.3.9.
(See Note A-9.36.2.8.(1).)

Tables 9.36.2.8.-A and B

Table 9.36.2.8.-A
Effective Thermal Resistance of Assemblies Below-Grade or in Contact with the Ground in Building without a Heat-Recovery Ventilator
Forming Part of Sentences 9.36.2.8.(1) to (9)

Building Assembly Below-*Grade* or in Contact with the Ground [1]	Heating Degree-Days of *Building* Location,[2] in Celsius Degree-Days					
	Zone 4 < 3000	Zone 5 3000 to 3999	Zone 6 4000 to 4999	Zone 7A 5000 to 5999	Zone 7B 6000 to 6999	Zone 8 ≥ 7000
	Minimum Effective Thermal Resistance (RSI), (m2×K)/W					
Heated floors[6]	2.32	2.32	2.32	2.84	2.84	2.84

Table 9.36.2.8.-B
Effective Thermal Resistance of Assemblies Below-Grade or in Contact with the Ground in Building with a Heat-Recovery Ventilator
Forming Part of Sentences 9.36.2.8.(1) to (9)

Building Assembly Below-*Grade* or in Contact with the Ground [1]	Heating Degree-Days of *Building* Location,[2] in Celsius Degree-Days					
	Zone 4 < 3000	Zone 5 3000 to 3999	Zone 6 4000 to 4999	Zone 7A 5000 to 5999	Zone 7B 6000 to 6999	Zone 8 ≥ 7000
	Minimum Effective Thermal Resistance (RSI), (m2×K)/W					
Heated floors[6]	2.32	2.32	2.32	2.84	2.84	2.84

Table 9.38.1.1.

Provision 9.36.2.8. Objective and Functional Statements

(1) [F92-OE1.1]

Division A, Article 2.2.1.1.

Division A, Article 2.2.1.1. Objectives

OE Environment

An objective of this Code is to limit the probability that, as a result of the design or construction of the building or facility, the environment will be affected in an unacceptable manner.

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OE1 Resources

An objective of this Code is to limit the probability that, as a result of the design or construction of the building or facility, resources will be used in a manner that will have an unacceptable effect on the environment. The risks of unacceptable effect on the environment due to use of resources addressed in this Code are those caused by –

OE1.1 – excessive use of energy

Division A, Article 2.2.1.1.

Division A, Article 3.2.1.1. Functional Statements

1) The objectives of this Code are achieved by measures, such as those described in the acceptable solutions in Division B, that are intended to allow the building or its elements to perform the following functions (see Note A-3.2.1.1.(1).:

F92 To limit the amount of uncontrolled thermal transfer through the building envelope.

Application

This variance applies to Part 9 buildings where a heating system creates a heated floor in contact with the ground, either fully below the frost line or above the frost line.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

1. A minimum RSI 1.32 (R 7.5) under slab insulation complies when installed for a heated floor in contact with the ground.

This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition.

Issued by the Provincial Building Administrator

[Original signed]

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